HW4

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Part 1.

- ## Governing PDE (undamped string)

The mode shapes are determined from the **undamped free vibration problem**.
For a uniform string of length L, tension T, and mass density ρ:

$$\rho \frac{\partial^2 w(x,t)}{\partial t^2} = T \frac{\partial^2 w(x,t)}{\partial x^2}, 0 < x < L$$

with fixed boundary conditions

$$w(0,t) = 0, w(L,t) = 0$$

- ## Separation of variables

Assume a separable solution:

$$\boxed{w(x,t) = X(x)\,T(t)}$$

Substitute into the PDE:

$$\rho X(x)\ddot{T}(t) = T\,X''(x)T(t)$$

Divide both sides by $X(x)T(t)$:

$$\frac{\ddot{T}}{T} = \frac{T}{\rho}\frac{X''}{X}$$

Since the left side depends only on t and the right side only on x, both must equal a **separation constant**, say $-\omega^2$.

- ## Resulting eigenvalue problems

Time equation

$$\boxed{\ddot{T} + \omega^2 T = 0}$$

Spatial (mode shape) equation

$$\boxed{X''(x) + k^2 X(x) = 0} \text{ where } k^2 = \frac{\rho}{T}\omega^2$$

This is the **mode shape equation**.

- ## General solution of the spatial equation

The general solution is

$$X(x) = A\sin(kx) + B\cos(kx)$$

- ## Apply boundary conditions (fixed ends)

At $x = 0$

$$X(0) = 0 \Rightarrow B = 0$$

At $x = L$

$$X(L) = A\sin(kL) = 0$$

For nontrivial solutions ($A \neq 0$):

$\sin(kL) = 0 \Rightarrow kL = n\pi, n = 1,2,3, \ldots$

- ## Mode shapes and natural frequencies

Mode shapes

$$\boxed{\phi_n(x) = \sin\left(\frac{n\pi x}{L}\right)}$$

Natural frequencies

$$\boxed{\omega_n = \frac{n\pi}{L}\sqrt{\frac{T}{\rho}}}$$

These $\phi_n(x)$ are the **mode shapes** used in the modal expansion.

- ## Summary of mode shapes.

The mode shape equation comes from separation of variables applied to the undamped string PDI

$$\boxed{X''(x) + \left(\frac{n\pi}{L}\right)^2 X(x) = 0, X(0) = X(L) = 0}$$

leading to

$$\boxed{\phi_n(x) = \sin\left(\frac{n\pi x}{L}\right)}$$

- ## Inclusion of a point viscous damper

Suppose a **linear viscous damper** with damping coefficient c is attached at

$x = a \ (0 < a < L)$

The damper exerts a transverse force proportional to the **local transverse velocity**:

$$f_d(x, t) = -c\,\frac{\partial w(a, t)}{\partial t}\,\delta(x - a)$$

where $\delta(\cdot)$ is the Dirac delta function.

The governing equation becomes

$$\rho\,\frac{\partial^2 w}{\partial t^2} - T\,\frac{\partial^2 w}{\partial x^2} + c\,\delta(x - a)\,\frac{\partial w(a, t)}{\partial t} = 0$$

- ## Modal expansion

We expand the displacement in the **undamped normal modes** of the fixed–fixed string:

$$w(x, t) = \sum_{n=1}^{\infty} q_n(t)\,\phi_n(x)$$

where the mode shapes are

$$\phi_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

These satisfy:

$$\int_0^L \phi_n(x)\phi_m(x)\,dx = \frac{L}{2}\,\delta_{nm}$$

and

$$\omega_n = \frac{n\pi}{L}\sqrt{\frac{T}{\rho}}$$

- **Substitution and projection**

Differentiate:

$$\frac{\partial w}{\partial t} = \sum_n \dot{q}_n(t)\,\phi_n(x)$$

At the damper location:

$$\frac{\partial w(a,t)}{\partial t} = \sum_n \dot{q}_n(t)\,\phi_n(a)$$

Substitute the modal expansion into the PDE and project onto the m-th mode by multiplying by $\phi_m(x)$ and integrating from 0 to L.

- **Resulting modal equations**

After using orthogonality, the modal equations become:

$$\boxed{\ddot{q}_m(t) + \omega_m^2\,q_m(t) + \sum_{n=1}^{\infty} C_{mn}\,\dot{q}_n(t) = 0}$$

where the **modal damping coupling coefficients** are

$$\boxed{C_{mn} = \frac{2c}{\rho L}\,\phi_m(a)\,\phi_n(a) = \frac{2c}{\rho L}\,\sin\left(\frac{m\pi a}{L}\right)\sin\left(\frac{n\pi a}{L}\right)}$$

- **Key observations**

  (a) Damping is non-proportional

The damping matrix $\mathbf{C} = [C_{mn}]$ is **dense**, not diagonal.
Modes are coupled through damping, even though the undamped system is diagonal.

  (b) Location dependence

The effectiveness of damping on mode n depends on

$$\sin\left(\frac{n\pi a}{L}\right)$$

If the damper is placed at a **node** of a particular mode, that mode is not directly damped.

  (c) Matrix form

Truncating to N modes:

$$\ddot{\mathbf{q}} + \mathbf{C}\,\dot{\mathbf{q}} + \mathbf{K}\,\mathbf{q} = \mathbf{0}$$

with

$$\mathbf{K} = \mathrm{diag}\left(\omega_1^2, \ldots, \omega_N^2\right), \mathbf{C}_{mn} = \frac{2c}{\rho L}\phi_m(a)\phi_n(a)$$

- **Final result**

$$\boxed{\ddot{q}_m + \omega_m^2 q_m + \frac{2c}{\rho L}\sum_{n=1}^{\infty} \sin\left(\frac{m\pi a}{L}\right)\sin\left(\frac{n\pi a}{L}\right)\dot{q}_n = 0}$$

Part 2.

Derivation of the **single-mode nonlinear modal equation** for a **pinned–pinned uniform beam** subjected to a **harmonic point force at mid-span**, including the **first nonlinear term in curvature** (finite but small rotations).
The final result is a **forced Duffing oscillator** for the first mode.

- ## Physical model and assumptions
  - Beam length: L
  - Flexural rigidity: EI
  - Mass per unit length: ρA
  - Pinned–pinned boundary conditions:
    $w(0,t) = w(L,t) = 0, w''(0,t) = w''(L,t) = 0$
  - External force:
    $f(x,t) = F_0\cos(\Omega t)\,\delta\left(x - \frac{L}{2}\right)$
  - **Euler–Bernoulli beam theory**
  - **Large displacement–small rotation assumption**, keeping the *first nonlinear curvature term*
- ## Nonlinear curvature approximation

Exact curvature:

$$\kappa = \frac{w''}{(1 + w'^2)^{3/2}}$$

For small but finite rotations ($|w'| \ll 1$), expand:

$$\boxed{\kappa \approx w'' - \frac{3}{2}w'^2 w''}$$

- ## Governing equation

Using Hamilton's principle or force balance, the nonlinear PDE becomes:
Kinetic energy:

$$\boxed{T = \frac{1}{2}\int_0^L \rho\, A\, \dot{w}^2\, dx}$$

Bending strain energy:

$$U = \frac{1}{2}\int_0^L E\, I\kappa^2\, dx$$

Substitute the approximation:

$$\kappa^2 \approx \left(w''\right)^2 - 3w'^2\left(w''\right)^2$$

Neglect higher-order terms (w'^4):

$$\boxed{U = \frac{1}{2}EI\int_0^L (w'')^2 dx - \frac{3}{2}EI\int_0^L w'^2\left(w''\right)^2 dx}$$

The first term is linear bending; the second term introduces **cubic geometric nonlinearity**.
Virtual work:

$$\boxed{\delta W = \int_0^L f\,(x,t)\,\delta w\, dx = F_0\cos(\Omega t)\,\delta w\left(\frac{L}{2}, t\right)}$$

$$\rho A\,\ddot{w} + EI\,w'''' - \frac{3}{2}EI\left(w'^2 w''\right)'' = F_0\cos(\Omega t)\,\delta\left(x - \frac{L}{2}\right)$$

- **Single-mode expansion**

Using a **single-mode approximation**:

$$\boxed{w(x,t) = q(t)\,\phi_1(x)}$$

For a pinned–pinned beam, the linear mode shapes are

$$\phi_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

so

$$\phi_1(x) = \sin\left(\frac{\pi x}{L}\right)$$

- **Galerkin projection**

Substitute into the PDE, multiply by $\phi_1(x)$, and integrate over $x \in [0,L]$.

Kinetic energy

$$T = \frac{1}{2}\dot{q}^2\rho A\int_0^L \phi_1^2\,dx = \frac{1}{2}M_1\dot{q}^2$$

$$\boxed{M_1 = \rho A\int_0^L \phi_1^2\,dx = \frac{\rho AL}{2}}$$

Linear strain energy

$$U_L = \frac{1}{2}EIq^2\int_0^L (\phi_1'')^2 dx$$

Since

$$\phi_1'' = -\frac{\pi^2}{L^2}\phi_1$$

$$\boxed{U_L = \frac{1}{2}K_1 q^2, K_1 = \frac{EI\pi^4}{2L^3}}$$

Nonlinear strain energy

$$U_{NL} = \frac{3}{2}EIq^4\int_0^L \phi_1'^2\,\phi_1''^2\,dx$$

Compute derivatives:

$$\phi_1' = \frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right), \phi_1'' = -\frac{\pi^2}{L^2}\sin\left(\frac{\pi x}{L}\right)$$

Evaluation of the integral yields:

$$\boxed{U_{NL} = \frac{1}{4}\alpha q^4, \alpha = \frac{3EI\pi^6}{16L^5}}$$

External work

$$W = F_0 q(t)\phi_1\left(\frac{L}{2}\right)\cos(\Omega t) = F_0 q(t)\cos(\Omega t)$$

since $\phi_1(L/2) = 1$.

Reduced Lagrangian

$$\boxed{\mathcal{L} = \frac{1}{2}M_1\dot{q}^2 - \frac{1}{2}K_1 q^2 - \frac{1}{4}\alpha q^4 + F_0 q\cos(\Omega t)}$$

Euler–Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}}\right) - \frac{\partial\mathcal{L}}{\partial q} = 0$$

This yields:

$$\boxed{M_1\ddot{q} + K_1 q + \alpha q^3 = F_0\cos(\Omega t)}$$

- **Final single-mode nonlinear equation**

Combining all terms:

$$\boxed{M_1\ddot{q} + K_1 q + \alpha q^3 = F_0\cos(\Omega t)}$$

or explicitly:

$$\boxed{\frac{\rho AL}{2}\,\ddot{q} + \frac{EI\pi^4}{2L^3}\,q + \frac{3EI\pi^6}{16L^5}\,q^3 = F_0\cos(\Omega t)}$$

- **Standard Duffing form**

Dividing by M_1:

$$\boxed{\ddot{q} + \omega_1^2 q + \beta q^3 = f\cos(\Omega t)}$$

where

$$\omega_1^2 = \frac{EI\pi^4}{\rho AL^4}, \beta = \frac{3EI\pi^6}{8\rho AL^6}, f = \frac{2F_0}{\rho AL}$$